

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$P(t) = 260(1.04)^{(\frac{6}{4})t}$$

The function P models the population, in thousands, of a certain city t years after **2003**. According to the model, the population is predicted to increase by **4%** every n months. What is the value of n ?

Answer

- A. **8**
- B. **12**
- C. **18**
- D. **72**

Correct Answer: A

Rationale

Choice A is correct. It’s given that the function P models the population, in thousands, of a certain city t years after **2003**. The value of the base of the given exponential function, **1.04**, corresponds to an increase of **4%** for every increase of **1** in the exponent, $(\frac{6}{4})t$. If the exponent is equal to **0**, then $(\frac{6}{4})t = 0$. Multiplying both sides of this equation by $(\frac{4}{6})$ yields $t = 0$. If the exponent is equal to **1**, then $(\frac{6}{4})t = 1$. Multiplying both sides of this equation by $(\frac{4}{6})$ yields $t = \frac{4}{6}$, or $t = \frac{2}{3}$. Therefore, the population is predicted to increase by **4%** every $\frac{2}{3}$ of a year. It’s given that the population is predicted to increase by **4%** every n months. Since there are **12** months in a year, $\frac{2}{3}$ of a year is equivalent to $(\frac{2}{3})(12)$, or **8**, months. Therefore, the value of n is **8**.

Choice B is incorrect. This is the number of months in which the population is predicted to increase by **4%** according to the model $P(t) = 260(1.04)^t$, not $P(t) = 260(1.04)^{(\frac{6}{4})t}$.

Choice C is incorrect. This is the number of months in which the population is predicted to increase by **4%** according to the model $P(t) = 260(1.04)^{(\frac{4}{6})t}$, not $P(t) = 260(1.04)^{(\frac{6}{4})t}$.

Choice D is incorrect. This is the number of months in which the population is predicted to increase by **4%** according to the model $P(t) = 260(1.04)^{(\frac{1}{6})t}$, not $P(t) = 260(1.04)^{(\frac{6}{4})t}$.